BIODATA / ABOUT ME

ENGINEERING PORTFOLIO



AIZAT ZUL HAIRIE BIN WAHI
(AIZAT / JAT)

Six Sigma Yellow Belt

Google Certified Data Analyst

UTM Endowment Scholar

Kuala Paka, Terengganu, Malaysia | linkedin.com/in/aizat-zulhairie-961053295

PROFESSIONAL INTRODUCTION

My name is Aixat Zul Hairie, and I am currently pursuing a degree in Technology with Education, majoring in Mechanical Engineering with a specialization in Welding Technology at Universiti Teknologi Malaysia (UTM). My educational background is rooted in technical and mechanical engineering, which has equipped me with both theoretical knowledge and practical skills. The addition of Welding Technology as a minor complements my major by deepening my understanding of material behaviour, joining processes, and inspection methods, all of which are crucial in the field of quality control and quality assurance. With this combined expertise, I am particularly interested in ensuring product reliability and maintaining high standards in manufacturing and engineering processes.

ACADEMIC BACKGROUND

UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Bachelor of Technology with Education (Mechanical Engineering) (Hons.)
Minor in Welding Technology

Oct 2023 – Sep 2027

Kolej Matrikulasi Kejuruteraan Pahang

Pre-University, Mechanical Engineering

Oct 2022 – May 2023

Sekolah Menengah Teknik Terengganu (TEKTRA / SMTT)

Sijil Pelajaran Malaysia (SPM), Mechanical Engineering

Feb 2020 – Dec 2021

TECHNICAL COMPETENCIES

- Welding Technology & Joining Processes
- Quality Control & Quality Assurance (QC/QA)
- Data Analysis & Visualization (Power BI)
- Computer-Aided Design (AutoCAD / SolidWorks)
- Nondestructive Testing (NDT)
- Process Improvement (Six Sigma, DMAIC, 7QC Tools)

AREA OF INTEREST

- Quality Control / Quality Assurance (QC/QA)
- Welding Technology & Inspection
- Data Analytics & Process Improvement
- Manufacturing & Engineering Process

KEY SKILLS & RELATED PROJECTS

Microsoft Power BI

- Welding QA/QC Dashboard – Power BI Analytics
- Harnessing the Power of Data with Power BI
- Extract, Transform and Load Data in Power BI
- Data Modeling in Power BI

Quality Control

- EDU TEKTRA Mobile Apps
- Arc Welding Jigs & Training Features
- Six Sigma Yellow Belt Capstone Project: Reducing Weld Defects in Metal Piping Production
- Welding QA/QC Dashboard – Power BI Analytics
- Engineering Project Management Specialization
- Six Sigma Yellow Belt Specialization
- Quality Improvement and Management

Welding Inspection

- Welding Instructor at Vocational College
- Arc Welding Jigs & Training Features
- Self Project : Machine Learning-Based Prototype for Welding Porosity Defect Detection

AutoCAD Mechanical

- **Design for Welding Fabrication Project (e.g. jigs, fixtures, and training features)**

Nondestructive Testing (NDT)

- NON-DESTRUCTIVE TESTING II FULL COURSE | NDT COURSE
- SMAW - CS PLATE - 3G WELDER (AWS D1.1:2015)
- Welding Processes in Heavy Manufacturing & Quality Controls

EDUCATION & ACADEMIC ACHIEVEMENTS

ENGINEERING PORTFOLIO

EDUCATION BACKGROUND



Universiti Teknologi Malaysia (UTM)

Bachelor of Technology with Education (Mechanical Engineering) (Hons.)
Minor in Welding Technology

Oct 2023 – Sep 2027

Current CGPA
3.99 (6/8)



Kolej Matrikulasi Kejuruteraan Pahang | Pahang Engineering Matriculation College (KMKPH)

Pre-University, Mechanical Engineering

Oct 2022 – May 2023

CGPA
4.00

Activities & Societies:
- Peer Assisted Learning (Mechanical Engineering)

Skills:
- Computer-Aided Design (CAD)
- Microsoft Excel



Sekolah Menengah Teknik Terengganu (TEKTRA / SMTT)

Sijil Pelajaran Malaysia (SPM), Mechanical Engineering

Feb 2020 – Dec 2021

Result
7A3B

Activities & Societies:
- Assistant Developer of EDU TEKTRA Project
- School Prefect
- TEKTRA Cyber Bridge Prefect

ACADEMIC ACHIEVEMENTS & AWARDS



ENDOWMENT MERDEKA SCHOLARSHIP AWARDS

Jul 2024 | Universiti Teknologi Malaysia (UTM)



DIRECTOR'S AWARDS

Jul 2023 | Pahang Engineering Matriculation College (KMKPH)



EXCELLENT STUDENTS AWARDS SPM 2021

Jul 2022 | Sekolah Menengah Teknik Terengganu (TEKTRA / SMTT)



INTERNATIONAL COCURICULUM AWARDS 2021

Aug 2021 | Sekolah Menengah Teknik Terengganu (TEXTRA / SMTT)



MALAYSIA YOUNG SCIENTIST CONFERENCE AND EXHIBITION 2021 (BRONZE AWARDS)

Apr 2021 | Sekolah Menengah Teknik Terengganu (TEKTRA / SMTT) & Universiti Teknologi Malaysia (UTM)

INTERNSHIP EXPERIENCE

ENGINEERING PORTFOLIO

INTERNSHIP 1

Insulet®

MANUFACTURING ENGINEER (MAINTENANCE AND PROCESS)

INTERNSHIP



DURATION
6 JULY 2026 – 5 OCTOBER 2026
(3 MONTHS)

COMPANY
INSULET CORPORATION
(MALAYSIA)





ROLE & KEY CONTRIBUTIONS

- Attached to the Process Engineering Team, supporting process improvement, quality investigations, and engineering studies within a regulated medical device manufacturing environment.
- Closed 15+ NCMR investigations using DMAIC, 5 Why Analysis, Fishbone Diagram, process mapping, and station verification techniques.
- Assisted in PFMEA remediation and PFMEA-DFMEA alignment activities to strengthen risk management and process controls.
- Conducted Ppk/Cpk analysis and statistical studies using Minitab to evaluate process capability and performance.
- Identified Critical-to-Process (CTP) contact points and supported manufacturing process optimisation initiatives.
- Collaborated with cross-functional engineering and quality teams to drive root cause investigations and continuous improvement projects.
- Supported maintenance and production activities while maintaining compliance with Good Documentation Practices (GDP) and current Good Manufacturing Practices (cGMP), ensuring accurate documentation, traceability, and adherence to established procedures.

KEY TOOLS & METHODOLOGIES

DMAIC | 5 WHY ANALYSIS | FISHBONE DIAGRAM | PROCESS MAPPING | PFMEA / DFMEA | Ppk/Cpk (Minitab) | STATION VERIFICATION | GDP / cGMP

INTERNSHIP 2

KOLEJ VOKASIONAL
KEMENTERIAN PENDIDIKAN MALAYSIA

WELDING INSTRUCTOR

INTERNSHIP



DURATION
MARCH 2026 – JUNE 2026
(4 MONTHS)

LOCATION
VOCATIONAL COLLEGE
KUANTAN, PAHANG, MALAYSIA



ROLE & KEY CONTRIBUTIONS

- Assisted in delivering theoretical and practical training for Sijil Vokasional Malaysia (SVM) and Diploma Vokasional Malaysia (DVM) students in welding technology.
- Supported teaching of welding inspection, welding supervision, welding processes, and welding quality control aligned with industry practices.
- Demonstrated welding techniques and proper use of SMAW, GMAW, and GTAW equipment during practical workshop sessions.
- Guided students in identifying welding defects and imperfections and understanding basic inspection principles used in fabrication and structural welding.
- Assisted in supervising practical assessments while ensuring compliance with welding procedures, safety standards, and workshop best practices.
- Contributed to competency-based training that reflects real industrial welding environments and fabrication workflows.



KEY TOOLS & METHODOLOGIES

SMAW (Stick) | GMAW (MIG) | GTAW (TIG) | WELDING INSPECTION | QUALITY CONTROL | WELDING SUPERVISION | SAFETY STANDARDS | FABRICATION WORKFLOW

SKILLS & CERTIFICATIONS

ENGINEERING PORTFOLIO

CORE SKILLS OVERVIEW

Quality & PLM
NCMR investigation (15+ cases), root cause analysis (5 Why, Fishbone, DMAIC), process mapping, PFMEA / DFMEA, Arena PLM.

Data Analytics
Data analysis, visualization and reporting using Power BI, advanced Excel (formulas, Pivot, analysis), and data-driven decision making.

Data Engineering & Coding
Building ETL pipelines, data transformation and automation using Apache Spark, dbt, GitHub Codespaces, HTML and CSS.

Machine Learning
Developed a welding defect (porosity) detection model using Google's machine learning platform (image classification).

Manufacturing & CAD
Certified 3G SMAW welder (CIDB), welding process and inspection, NDT (PT), and thermal analysis using SOLIDWORKS Simulation.

Office Productivity
Proficient in Microsoft Office suite (Word, Excel, PowerPoint).

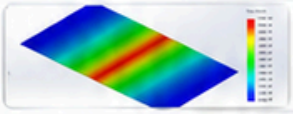
TECHNICAL TOOLS & SOFTWARE

- **SOLIDWORKS** (CAD & Simulation)
- **Power BI** (Data Visualization)
- **Arena** (Arena PLM) (NCMR / Quality)
- **Python** (Data & ML)
- **Apache Spark** (Apache Spark) (Data Engineering)
- **dbt** (dbt) (Data Transformation)
- **Apache Airflow** (Workflow Automation)
- **Microsoft Excel** (Data Analysis)
- **HTML** (HTML)
- **CSS** (CSS)

SELECTED ENGINEERING PROJECTS



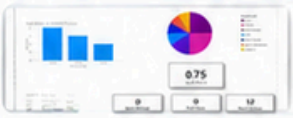
MI-Based Welding Porosity Detector
Developed an image classification model to detect welding porosity and automate visual inspection.



Gas Welding Lab – Material & Thermal Analysis
Analyzed material suitability and heat distribution for gas welding using SOLIDWORKS Simulation.



End-to-End Data Pipeline: NYC Yellow Taxi Analytics (Q1 2024)
Built an automated ELT pipeline (DuckDB, dbt, Airflow) processing 9.5M+ taxi records with data quality tests.



Welding QA/QC Dashboard – Power BI
Developed an interactive dashboard to monitor welding defects, trends and welder performance for QA/QC.

CERTIFICATIONS & PROFESSIONAL CREDENTIALS

Basics of Robotics
Siemens
Issued: Oct 2026

Understanding Digital Transformation
MyDIGITAL Corporation
Issued: Oct 2026 | ID: 6KKPYC07AG7S

Google Project Management Professional Certificate (v.3)
Google
Issued: Aug 2026

Arena Basic User/Approver Training Module
Insulet Corporation
Issued: Jul 2026 | ID: 543854

Google Data Analytics Professional Certificate (v.3)
Google
Issued: Jun 2026 | Skills: Data Analytics, PL/SQL, +4

Upskilling Program: Cloud Computing Essentials
Universiti Teknologi Malaysia
Issued: Feb 2026 | Skills: Microsoft Azure, Cloud Computing

Welding Processes in Heavy Manufacturing & Quality Controls
L&T EduTech
Issued: May 2025 | ID: 9060M8B7VQL1 | Skills: NDT

SMAW - CS PLATE - 3G WELDER (AWS D1.1:2015)
CIDB Technologies Sdn Bhd
Issued: Aug 2025 | Expires: Aug 2028
ID: BE/MY/TR/25/WAB3/1663959 | Skills: NDT

Non-Destructive Testing II Full Course | NDT Course
Augmintech Education Pvt. Ltd.
Issued: Sep 2025 | ID: UC-bf6f16f8-7f37-4efd-943a-648cc84ad4f4

Heavy Manufacturing
L&T EduTech
Issued: May 2025 | ID: GTF4SASHZZ4S
Skills: Piping and Instrumentation Drawing (P&ID)

Data Modeling in Power BI
Microsoft
Issued: May 2025 | ID: L4CRALU1W2IJ

Extract, Transform and Load Data in Power BI
Microsoft
Issued: May 2025 | ID: P2JSG24KZOC5

Harnessing the Power of Data with Power BI
Microsoft
Issued: May 2025 | ID: 63J2JH6D3PJL

Preparing Data for Analysis with Microsoft Excel
Microsoft
Issued: May 2025 | ID: P7UNGMSR9YPM

Quality Improvement and Management
Board Infinity
Issued: May 2025 | ID: 66591M1JC5OF
Skills: Quality Control, Quality Assurance

Six Sigma Yellow Belt Specialization
Kennesaw State University
Issued: May 2025 | ID: UJN49202WT9O
Skills: Quality Control, Quality Assurance

Engineering Project Management Specialization
Rice University
Issued: May 2025 | ID: JUEM22V9O6AW
Skills: Engineering Management, Quality Control, +1

VOLUNTEERING & CO-CURRICULAR ACTIVITIES

ENGINEERING PORTFOLIO

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FEST JOHOR – TECHNICAL TEAM MEMBER

Yayasan Bank Rakyat | Jun 2024 (1 Month) | Technical Team Member

- Served as a Technical Team Member and volunteer at FEST Johor, working with TVET students from various vocational colleges.
- Supported STEM activities and the development of industry-relevant technical skills.
- Gained experience in technical support, knowledge sharing, and cross-team collaboration.

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HARI KOMUNITI 2026 – VOLUNTEER STAFF

PLUS Malaysia | Jan 2026 (1 Month) | Volunteer Staff

- Participated in Hari Komuniti 2026 at Masjid Kahar, Pontian, Johor, representing PLUS Section 4.
- Contributed to community engagement and social development activities.
- A meaningful experience in giving back to the community and strengthening teamwork with fellow volunteers.

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INTERNATIONAL EXCHANGE – FACILITATOR

UTM Universiti Teknologi Malaysia | Oct 2025 (1 Month) | Facilitator

- Served as a facilitator for an international exchange programme with students from Kaichi Mirai High School, Japan (20 October 2025).
- Led discussions on SNS digitalization etiquette between Japan and Malaysia.
- Bridged cultural and language differences through teamwork and mutual respect.
- Gained broader perspectives on cross-cultural communication, digital awareness, and cultural intelligence in the era of Industry 4.0.

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UTM GAMES – RUGBY PLAYER

KOLEJ 10 UTM | Nov 2024 (1 Month) | Rugby Player

- Represented Kolej 10 in the UTM Games rugby competition among various colleges.
- Strengthened teamwork, discipline, resilience, and leadership on and off the field.
- Gained valuable experience in teamwork, time management, and sportsmanship.

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PERSATUAN MAHASISWA ANAK TERENGGANU – EXCO

Persatuan Mahasiswa Anak Terengganu UTM (PERMATA UTM) | 2024 – 2025 (1 Year) | EXCO Bureau & Public Relation (PR)

- Served as an EXCO member for the Bureau and Public Relation (PR) at PERMATA UTM.
- Managed club communications, event promotion, and stakeholder engagement.
- Supported the planning and execution of student programmes that strengthen the Terengganu student community at UTM.

ENGINEERING PROJECTS

ENGINEERING PORTFOLIO

PROJECT HIGHLIGHTS (1/2)

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PIPING DRAWINGS (SELF PROJECT)

Dec 2025 – Present | UTM | SOLIDWORKS, CAD and +1 skill

- Practice developing Piping and Instrumentation Diagrams (P&ID) using SolidWorks.
- Interpret engineering standards, process flow and key components (valves, instruments, piping routes, safety devices).
- Translate P&ID symbols into accurate 2D/3D representations and improve design workflow for industrial plant documentation.

INSERT PROJECT PHOTO HERE

MACHINE LEARNING-BASED PROTOTYPE FOR WELDING POROSITY DEFECT DETECTION

Feb 2026 – Feb 2026 | UTM | Machine Learning, HTML and +2 skills

- Developed an automated feasibility study for welding porosity defect detection using Google's machine learning platform.
- Designed an image classification model with a standardized workflow: Image Input → Pre-processing → ML Classification → Output Display.
- Future roadmap: real-time camera detection, multi-defect classification (e.g. cracks, slag inclusion), and higher accuracy with industrial datasets.

INSERT PROJECT PHOTO HERE

GAS WELDING PROJECT LAB – MATERIAL SELECTION & THERMAL ANALYSIS

Dec 2025 – Dec 2025 | UTM | Thermal Analysis, SOLIDWORKS and +2 skills

- Investigated various metals to determine suitable materials for gas welding.
- Conducted thermal simulations using SolidWorks to analyze heat distribution.
- Enhanced understanding of material behavior under high temperatures and welding processes.

INSERT PROJECT PHOTO HERE

SIX SIGMA YELLOW BELT CAPSTONE PROJECT: REDUCING WELD DEFECTS IN METAL PIPING PRODUCTION

May 2025 – May 2025 | UTM | Six Sigma, DMAIC and +5 skills

- Applied the DMAIC methodology to reduce weld defects in a metal piping production line.
- Identified a baseline defect rate of 12.4% (above 5% threshold) and defined CTQs (defect rate, rework hours, first-pass yield).
- Analyzed process variability, identified root causes, and proposed improvement actions (process control and operator training).
- Target: defect rate < 5%, rework hours –40%, and first-pass yield ≥ 95%.

INSERT PROJECT PHOTO HERE

ARC WELDING JIGS & TRAINING FEATURES

– | UTM | Arc Welding, AutoCAD and +5 skills

- Designed, modelled and fabricated custom welding jigs and training features.
- Produced detailed CAD models and technical drawings using SolidWorks.
- Developed adjustable jigs for real-world welding simulation with ergonomic and safe setup.
- Improved joint alignment accuracy and welding angle consistency.
- Aligned with TVET and industry competency standards for welder training.

ENGINEERING PROJECTS

ENGINEERING PORTFOLIO

PROJECT HIGHLIGHTS (2/2)

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END-TO-END MODERN DATA PIPELINE: NYC YELLOW TAXI ANALYTICS (Q1 2024)

Q1 2024 | UTM | Contributors: Muhammad Hafiy

- Built an automated, production-grade ELT data pipeline processing over 9.5 million NYC yellow taxi records.
- Ingested raw Parquet and CSV files into DuckDB and applied a 4-step data cleaning and validation funnel using dbt models.
- Structured data into a high-performance star schema (fct_trips and dim_zones).
- Automated the workflow with Apache Airflow with dbt data quality tests (zero-defect reporting).

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WELDING QA/QC DASHBOARD – POWER BI ANALYTICS

– | UTM | Quality Control, Microsoft Power BI and +2 skills

- Developed an interactive Power BI dashboard to monitor and analyze welding defects in fabrication and pipeline projects.
- Features: KPI cards (inspections, defects, defect rate %, repairs), defect distribution by type and process, trend analysis, welder performance table, and project drill-downs.
- Provided data-driven insights to reduce rework and improve quality control.

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MIRABRUSH PROJECT

May 2023 – Jun 2023 | UTM | Quality Management, Critical Thinking and +1 skills

- Developed MiraBrush, a 2-in-1 toothbrush with an integrated toothpaste dispenser for students and frequent travelers.
- Served as Product Designer and Quality Checker, applying the DMAIC methodology.
- Designed the ergonomic structure, selected materials, and resolved a leakage issue in the internal tube mechanism.
- Delivered a more practical, reliable and user-centered product design.

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EDU TEKTRA MOBILE APPS

Feb 2021 – Apr 2021 | SMTT (TEKTRA) | Contributors: Muhammad Nabil Hakem and Ahmad | Quality Control and Figma (Software)

- Served as Developer Assistant and Designer for the Edu Tektra mobile app.
- Developed a digital textbook hub for students and a room booking system for teachers (science labs, computer labs, workshops).
- Contributed to front-end implementation and designed a user-friendly interface to ensure smooth navigation.
- Improved academic access and room scheduling efficiency within the school environment.